

Shadman Salif Swanan

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Research Interests

Natural Language Processing, Large Language Models, Computer Vision, OCR, Deep Learning, Semantic Drift Analysis

Technical Skills

Programming Languages: Python, C++, JavaScript, PHP

Data Analysis: MySQL, Power BI, Microsoft Excel (Advanced), Process Modeling, EDA

AI/ML Frameworks: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Tensorflow, Hugging Face

Web & Tools: React, Vite, Node.js, Express.js, MongoDB, Git & GitHub, Tailwind CSS, Figma

Education

BRAC University	2022 – 2026
Bachelor of Science in Computer Science and Engineering	CGPA: 3.83 / 4.00 (Merit Based Scholarship)
Notre Dame College	2018 – 2021
Higher Secondary Certificate (Science)	GPA: 5.00 / 5.00
Ideal School and College	2008 – 2018
Secondary School Certificate (Science)	GPA: 5.00 / 5.00 (General Scholarship/ Dhaka Board)

Research Experience

Undergraduate Thesis Research BRAC University	2025 – 2026
– Conducting research on semantic drift in Bengali using diachronic word embeddings and transformer representations. Built OCR pipelines and fine-tuned Historical BanglaBERT on a 120-year corpus.	
Leaf Disease Research Independent	Apr 2026
– Developed TinyResNet, achieving 96.59% accuracy with distortion-aware augmentation techniques.	
Real-Time Fraud Detection System Independent	May 2026
– Built a fraud detection pipeline using LightGBM and developed a real-time transaction simulation console.	

Projects

- Shikkha.com** | React, Vite, Node.js, Express.js, MongoDB, JWT, Stripe, Tailwind CSS
- Developed a full-stack EdTech platform supporting course enrollment, online exams with automated grading, educator appointments, and e-commerce for books. Integrated JWT-based authentication, role-based access control, and Stripe Checkout.
- Predictive Telecom Churn Optimization Framework** | Python, Scikit-learn, HistGradientBoosting, Pandas, EDA, Feature Engineering
- Engineered a machine learning pipeline processing 100K customer logs with an advanced Gradient Boosting model, achieving a 0.6900 ROC-AUC score and 64.00% churn recall to project an estimated 2.22M USD in annualized retention savings.
- Token-Level POS Tagging Using RNN Models** | Python, TensorFlow, RNN, LSTM, GRU, BiLSTM, NLP
- Implemented and compared RNN, LSTM, GRU, and BiLSTM architectures for token-level part-of-speech tagging, evaluating sequence labeling performance across multiple recurrent neural network models.
- Disease Prediction Using Machine Learning** | Python, Scikit-learn, Pandas, NumPy
- Built KNN, Decision Tree, and Naive Bayes classifiers to predict diabetes from clinical features and compared model performance using standard classification metrics.
- Federal Flight Agency Mission-Critical Aerospace Network Design** | Cisco Packet Tracer, RIPV2, Static Routing, VLSM
- Designed a high-availability network for autonomous aircraft operations using VLSM and a hybrid routing strategy combining RIPV2 and static routes to ensure automatic failover and reliable communication.

Experience & Certifications

GCI World Program	Apr 2026 – Present
The University of Tokyo (Matsuo-Iwasawa Laboratory)	Remote
– Participating in an international AI education program focused on machine learning and data science.	
UX Design Decision-Making with AI & Design System Thinking with AI — GP Academy, 2026	
Conference Management Officer — Notre Dame College Model United Nations, 2019	

Co-Curricular Activities

Vice President, Photography Dept, NDNCS | Champion, BRACU Intra-Spring 2022 Football Tournament.
Runners-Up in Crusade 2026, RS Football Tournament | Completed Stage Drama Course (BRACU RS).

References

Dr. Chowdhury Mofizur Rahman
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